

Question 1

What is Cyber security? Explain the Concept of CIA triad in detail

Answer

Definition of Cybersecurity – The process of protecting the computing systems and the data inside it or the data it can access.

CIA triad commonly an anagram to remember the three important components of a system whose cybersecurity is ensured

1. **Confidentiality** – Only those who “should” access the information must be able to do it
2. **Integrity** – The Data stored is “immutable” which means it cannot be tampered hence ensuring that the data is reliable
3. **Availability** – The stored information is available when it is needed

Question 2

What are the requirements at OEM level and supplier level? Explain in detail

Cybersecurity in the Automotive sector is a shared responsibility between OEMs and their suppliers.

Cybersecurity Requirements at the OEM level

1. Strategic Integration of Cybersecurity – Security by Design
 - a. The Security by design approach implies that the OEMs must incorporate cybersecurity considerations from the initial design phases of the vehicle. This ensures that potential vulnerabilities are addressed proactively, reducing risks and costs at later stages.
 - b. OEMs also need to include defining security principles for hardware and software modules and assigning clear security roles.
2. Compliance with International Standards and Regulations
 - a. OEMs are required to manage cybersecurity activities across all phases of the vehicle lifecycle and rigorously analyze risk factors to meet the cybersecurity requirements set by ISO/SAE 21434
 - b. According to UNECE WP.29 OEMs must possess a certified Cybersecurity Management System (CSMS) and ensure that vehicles receive approval based on the stringent cybersecurity criteria. It also places demand on Secure OTA SW updates, addressing potential vulnerabilities in remote updates

3. Risk Assessment and Management
 - a. OEMs must conduct comprehensive threat, risk and hazard analysis (via the HARA and TARA methodologies) to identify potential cyberattacks and associated risks.
 - b. This Involves understanding various attack models, their motivations, and the potential impact (e.g., Data Theft, DoS, Intentional Manipulation and Unauthorized physical access)
4. Robust System Architecture:
 - a. OEMS need to implement a multi-layered cybersecurity approach, including perimeter security (protecting external connections line OBD ports, Wi-Fi, Bluetooth etc.), network security via firewalls, central gateways.
 - b. The ECU also must be protected against various attack classes through countermeasures like encryption, authentication, code analysis and physical tapering detection

Cybersecurity Requirements at the Supplier Level

1. Adherence to OEM specifications and Industry Standards – Suppliers must strictly adhere to cybersecurity specs provided by the OEM and comply with relevant industry standards like the ISO/SAE 21434.
2. Secure Development of Components and Software
 - a. AUTOSAR Implementation: Suppliers developing AUTOSAR-based software must ensure their implementations are secure, even if OEMs define the specific cryptographic algorithms. This includes secure coding practices and thorough testing.
 - b. Hardware Security: Suppliers providing hardware components like ECUs or Hardware Security Modules (HSMs) must ensure these components are designed with robust security features to withstand physical and cyberattacks (e.g., tamper detection, secure boot capabilities, cryptographic accelerators).
 - c. Vulnerability Management: Suppliers are responsible for identifying and mitigating vulnerabilities in their components and software before delivery to the OEM.
3. Supply Chain Cybersecurity - Suppliers are an integral part of the automotive supply chain and, as such, must implement their own robust cybersecurity management systems. This includes protecting their internal networks, intellectual property, and ensuring the security of their development and production processes to prevent the introduction of malicious code or compromised components into the supply chain.
4. Collaboration and Information Sharing - Suppliers need to work effectively with OEMs to integrate and validate end-to-end cybersecurity solutions. This often

involves sharing information about potential vulnerabilities, participating in joint threat modeling exercises, and collaborating on incident response.

5. **Bridging Knowledge Gaps** - Suppliers, especially those involved in new technologies like V2X, need to continuously update their technical knowledge in cybersecurity to keep pace with evolving threats and technological advancements. This ensures they can provide cutting-edge and secure solutions.

Question 3

What is the concept of Cyber security? explain the concept of CIA & AAA in detail

Cybersecurity, in the context of the automotive industry, refers to the protection of vehicle systems, networks, programs, devices, and data from cyberattacks, damage, or unauthorized access.

CIA Triad: Confidentiality, Integrity, Availability

The CIA triad is a foundational model for information security, outlining the three primary goals of any security system.

1. **Confidentiality** ensures that information is accessible only to authorized individuals, entities, or processes. It prevents unauthorized disclosure of sensitive data.
2. **Integrity** ensures that information is accurate, complete, and has not been modified or tampered with by unauthorized parties. It also implies that the data is consistent and trustworthy.
3. **Availability** ensures that authorized users can access information and resources when needed. It means that systems and data are operational and accessible without undue delay or interruption.

AAA: Authentication, Authorization, Accountable

AAA is a framework for controlling access to computer resources, enforcing policies, auditing usage, and providing information necessary to bill for services.

1. **Authentication**: is the process of verifying the identity of a user, device, or system.
2. **Authorization** is the process of determining what an authenticated user, device or system is permitted to do or access.
3. **Accountability** is process if tracking and logging user activities, system events, and resource consumption.

Question 4

Explain the below terms in detail:

4a. Cryptography

Answer

Cryptography is the process in which the information is “scrambled” so that certain pieces of the information is hidden

4b. Encryption

Answer

Encryption is the process of converting a message from one system of communication to another.

4c. Decryption

Answer

The process of converting a coded message back to its plain text form

Question 5

Write short note on the following related to vehicle network communication

5a. LIN (Local Interconnection Network)

Answer

LIN is a communication protocol which is primarily used for simple networking tasks, primarily due to it being cost effective. Unlike CAN it adopts a Serial Bus System, hence lacks the high speed and the complex architecture available on CAN bus. It's primarily used in Body electronics components like Power Windows, Seat Adjustments and Windshield wipers etc.,

LIN Architecture Components

LIN primarily adopts Master Slave architecture,

- the Master Node which typically is a Powerful ECU which is connected with a higher-level network like CAN, this acts as a gateway & controls the timing of all communication on the LIN bus
- The Slave Node are typically Simple Actuators, Switches or Intelligent sensors

Communication occurs over a single wire line, requiring simple wiring and hence the low cost.

The LIN protocol Data transfer rate is around 20 kbit/s, with the ability to be connected to a maximum of 16 subscribers (1 master + 15 slave nodes) which are typically connected in a Linear Bus Topology.

5b. MOST

MOST is a high-speed multimedia network technology specifically designed for use in vehicles. Its primary purpose is the efficient transmission of audio, video, and control data, making it a common choice for infotainment systems within a car. While MOST offers significant bandwidth for multimedia applications, the document highlights that its security features are often limited. This characteristic makes MOST a potential target for attackers who might seek to disrupt infotainment services or, if not properly secured, inject malicious data into other connected vehicle systems.

MOST Architecture

- MOST is designed to connect various multimedia devices within an automotive network.
- It allows for both synchronous and asynchronous data transmission, which is crucial for real-time audio and video.
- Communication typically occurs over Fiber optic cables, which contributes to its high speed and immunity to electromagnetic interference.
- Data Transfer Rate The data transfer rate of the MOST protocol can vary, but it is significantly higher than that of LIN, capable of supporting the data demands of high-quality audio and video.
- Topology MOST commonly adopts a ring topology, where data flows sequentially from one device to the next.

Question 6

Give the significance of cybersecurity in case of “V2I” type of communication of a vehicle with the help of different examples

V2I (Vehicle-to-Infrastructure) communication connects vehicles with road infrastructure like traffic lights, road signs, and smart parking systems. Cybersecurity is paramount here due to potential severe impacts on safety, efficiency, and privacy.

Safety: Compromising V2I can be life-threatening. For instance, an attacker manipulating traffic light timings (violating **Integrity**) could cause accidents. Similarly, sending false warnings or blocking real ones from Roadside Units (RSUs) (affecting **Availability** and **Integrity**) endangers drivers.

Efficiency: V2I optimizes traffic flow. A cyberattack disrupting adaptive traffic management systems (impacting **Availability** and **Integrity**) would lead to massive congestion and economic losses.

Privacy: V2I systems, like smart parking or electronic tolling, collect sensitive vehicle and driver data. Breaching this data's **Confidentiality** could expose personal travel patterns or financial information.

Robust cybersecurity, encompassing **Authentication, Authorization, Accounting**, and ensuring **Confidentiality, Integrity, and Availability**, is crucial for V2I's safe, reliable, and trustworthy operation, preventing chaos and maintaining public confidence in intelligent transportation systems.

Question 7

Explain the significant characteristics of the technologies below in terms of autonomous vehicles

7a. Positioning Technology

- This technology enables autonomous vehicles to know their precise location.
- It relies on **multi-sensor fusion**, combining **GNSS (GPS)** for global coordinates, **IMUs** for relative motion, **odometry**, and **HD Maps** for highly accurate localization.
- **Accuracy** (centimetres-level) is paramount for safe navigation and lane keeping. Systems prioritize **robustness and redundancy** to overcome signal loss (e.g., in tunnels).
- Cybersecurity is vital, as **GPS spoofing or jamming** could mislead the vehicle, compromising safety.

7b. Vision Technology

Vision technology allows autonomous vehicles to "see" and understand their surroundings. It integrates data from:

- **Cameras:** Provide rich visual data for object classification, lane detection, and traffic sign recognition, though sensitive to lighting.
- **LiDAR:** Generates precise 3D point clouds for accurate depth and obstacle detection, effective in varying light.
- **Radar:** Excels in adverse weather for long-range detection and velocity measurement. **Sensor fusion** combines these inputs for a comprehensive and robust environmental perception. Cybersecurity concerns include **adversarial attacks** that can trick AI perception or **jamming** of sensors, leading to dangerous misinterpretations.

Question 8

8a. What is Confidentiality in terms of in-vehicle communication networks (e.g., CAN bus)?

In terms of in-vehicle communication networks like the CAN bus, **Confidentiality** refers to ensuring that data transmitted across the network is **not disclosed to unauthorized entities**. While traditional CAN bus communication often prioritizes **Integrity** and **Availability** over Confidentiality (as data is typically exchanged between trusted ECUs within a closed system), it becomes crucial when:

- **Sensitive User Data:** Personal information (e.g., navigation history, biometric data, communication logs) is transmitted between ECUs, especially to/from infotainment systems.
- **Proprietary Information:** Critical vehicle control parameters or diagnostic data that could be exploited by competitors or malicious actors if intercepted.
- **External Interfaces:** When data from the CAN bus is bridged to external networks (e.g., telematics, V2X), confidentiality measures like encryption become essential to prevent eavesdropping and unauthorized access to internal vehicle data.

8b. Explain Integrity with an example related ECU firmware updates

Integrity ensures that data has not been altered or tampered with by unauthorized parties and remains accurate and complete. For **ECU firmware updates**, integrity is paramount:

Example: When an OEM releases a firmware update for a vehicle's Engine Control Unit (ECU), it's critical to ensure that the update file received by the vehicle is exactly what the OEM intended and hasn't been maliciously modified.

1. **Digital Signature:** The OEM digitally signs the firmware update package using their private key.
2. **Cryptographic Hash:** A cryptographic hash (e.g., SHA-256) of the update file is generated and included in the signed package.
3. **Verification:** Before installing the update, the vehicle's ECU uses the OEM's public key to verify the digital signature. It then calculates its own hash of the received update file and compares it to the hash provided in the signed package.
4. **Outcome:** If the signature is valid and the hashes match, the ECU confirms the **integrity** of the update – meaning it came from a trusted source and hasn't been altered. If they don't match, the update is rejected, preventing the

installation of potentially malicious or corrupted firmware that could compromise vehicle safety or functionality.

Question 9

What is Availability, and why is it critical for safety-critical automotive systems

Availability ensures that authorized users, devices, or systems can access information and resources when needed, without undue delay or interruption. It means that critical functions are operational and accessible continuously.

For safety-critical automotive systems, Availability is paramount because their uninterrupted operation directly impacts the safety of vehicle occupants and other road users.

- **Direct Safety Risk:** Systems like braking, steering, engine control, airbags, and Advanced Driver-Assistance Systems (ADAS) are designed to prevent accidents. If these systems become unavailable—perhaps due to a cyberattack (e.g., Denial-of-Service), a software glitch, or hardware failure—the driver could lose control, leading to severe accidents, injuries, or fatalities.
- **Real-time Operation:** Many safety-critical functions demand immediate, real-time responses. Even a momentary loss of availability can have catastrophic consequences, as the vehicle might fail to react appropriately to dynamic road conditions or emergencies.

Therefore, guaranteeing the highest level of Availability for these systems is a fundamental requirement, built through redundant designs, fault tolerance, and robust cybersecurity to prevent malicious disruption.

Question 10

List of two common cyber-attack surfaces in modern connected vehicles

1. **External Communication Interfaces:** These include technologies like V2X (Vehicle-to-Everything) communication, Wi-Fi, Bluetooth, and cellular (e.g., 5G). These interfaces provide direct remote access points to the vehicle's network, making them prime targets for attackers to intercept data, inject malicious commands, or exploit vulnerabilities in communication protocols.
2. **OBD-II Port (On-Board Diagnostics Port):** This physical port, typically located inside the vehicle, is used for diagnostics and maintenance. However, if not properly secured, it can serve as a direct gateway for attackers with physical

access to inject malicious code, reprogram ECUs, or extract sensitive data from the vehicle's internal communication buses (like CAN).

Question 11

What is an Electronic Control unit (ECU) and why is it a target for cyber-attacks?

An Electronic Control Unit (ECU) is an embedded computer within a vehicle, responsible for managing specific electrical systems or subsystems. Modern cars contain dozens, even hundreds, of ECUs controlling everything from the engine and brakes to airbags, steering, and infotainment. They process sensor data, execute software, and send commands to actuators, essentially acting as the vehicle's "brains."

ECUs are a prime target for cyber-attacks for several critical reasons:

1. **Control Over Critical Functions:** ECUs directly manage vital vehicle operations like acceleration, braking, and steering. Compromising an ECU grants attackers the ability to manipulate these functions, potentially causing dangerous situations, loss of control, or accidents.
2. **Gateway to Vehicle Networks:** Many ECUs are interconnected via internal communication networks (e.g., CAN bus). An attacker gaining control of one ECU can use it as a pivot to access and compromise other ECUs on the network, escalating their control over the entire vehicle system.
3. **Data Access:** ECUs process and store sensitive data, including vehicle performance metrics, diagnostic information, and sometimes personal user data. Attackers may target ECUs to steal this valuable information.
4. **Software Vulnerabilities:** Like any software-driven system, ECUs can have vulnerabilities that attackers exploit to gain unauthorized access, inject malicious code, or alter firmware.
5. **Remote Attack Vectors:** With increasing vehicle connectivity (Wi-Fi, Bluetooth, cellular, V2X), ECUs are exposed to remote attacks, making them accessible targets even without physical presence.

Question 12

Explain the term attack vector with an example in automotive cyber security.

An attack vector is the path or method an attacker uses to gain unauthorized access to a system or network, or to deliver a malicious payload. It's the specific entry point or vulnerability exploited.

Example in Automotive Cybersecurity: The Jeep Cherokee Hack

The 2015 Jeep Cherokee hack vividly demonstrated an attack vector.

- **Attack Vector:** The primary attack vector was the infotainment system's internet-facing cellular connection. This system, providing features like navigation and Wi-Fi, was directly accessible from the internet.
- **Explanation:** Security researchers exploited a vulnerability in the Uconnect software, gaining remote access to the infotainment system via its cellular modem. From there, they leveraged further vulnerabilities to bridge to the vehicle's internal CAN bus. This allowed them to remotely control critical functions like the air conditioning, radio, wipers, and, most alarmingly, the transmission and brakes while the vehicle was in motion.

Question 13

What is the difference between cybersecurity and functional safety in vehicles

1. Functional Safety:

- **Goal:** To prevent unreasonable risk arising from malfunctions of electrical and electronic (E/E) systems. It focuses on ensuring systems behave as intended and do not cause harm due to accidental hardware failures, systematic software errors, or environmental factors.
- **Nature of Threat:** Accidental, unintended failures or errors originating *within* the system itself.
- **Focus:** Preventing hazardous behaviour caused by system failures.
- **Standard:** Primarily governed by ISO 26262.
- **Example:** Ensuring a brake system activates correctly even if a sensor fails, or that software bugs don't cause unintended acceleration.

2. Cybersecurity:

- **Goal:** To protect vehicle systems, networks, and data from malicious attacks, unauthorized access, manipulation, or damage. It aims to safeguard the Confidentiality, Integrity, and Availability (CIA) of information and control systems.
- **Nature of Threat:** Intentional, external, and intelligent adversaries seeking to exploit vulnerabilities.
- **Focus:** Preventing harm caused by malicious intent.

- Standard: Primarily governed by ISO/SAE 21434 and UNECE WP.29.
- Example: Preventing a hacker from remotely disabling brakes, stealing personal data, or injecting malicious code into the vehicle's software.

Key Differences & Interdependency: **Functional safety addresses threats from internal system failures**, while **cybersecurity addresses threats from external malicious actors**. Functional safety deals with *unintended* hazardous behaviour, whereas cybersecurity deals with *intended* malicious actions. Crucially, a cybersecurity breach can directly *cause* a functional safety hazard (e.g., a cyberattack disabling the brakes). Therefore, robust cybersecurity is a prerequisite for achieving functional safety in modern connected and autonomous vehicles.

Question 14

Name two open-source tools that can be used for basic automotive cybersecurity experimentation and state their purpose.

1. CAN-utils is a suite of utilities for Linux that provides tools to interact with the Controller Area Network (CAN) bus. It allows users to sniff, send, and analyse CAN messages. For cybersecurity experimentation, it's fundamental for understanding normal CAN traffic, injecting custom messages to test ECU responses, or identifying potential vulnerabilities in how ECUs handle unexpected inputs.
2. Wireshark is general-purpose network protocol analyser, it is invaluable for automotive cybersecurity, especially with the rise of Automotive Ethernet. It allows users to capture and interactively browse the data flowing on a network. In an automotive context, it can be used to analyse Ethernet-based communication (e.g., between infotainment systems or ADAS sensors), or even CAN traffic if integrated with a suitable CAN interface, helping to identify unusual traffic patterns, unencrypted data, or potential attack vectors.

Question 15

Discuss the architecture of a modern connected vehicle and identify potential cybersecurity vulnerability in each subsystem.

A modern connected vehicle integrates numerous subsystems, communicating internally and externally, creating significant cybersecurity attack surfaces.

1. Telematics/External Communication Unit: (Cellular, Wi-Fi, V2X) This is the vehicle's primary interface to the outside world.

- Vulnerability: Remote unauthorized access via protocol exploits (e.g., in cellular modems) or V2X signal spoofing, deceiving the vehicle about its environment.
2. Infotainment System: (Navigation, media, smartphone integration) Often based on operating systems like Linux/Android.
 - Vulnerability: Malware injection via USB/Bluetooth, exploitation of app vulnerabilities, or serving as a pivot point to internal networks if not properly isolated.
 3. Gateway/Central ECU: Acts as a firewall and router, connecting internal networks (CAN, Ethernet) and controlling data flow.
 - Vulnerability: A single point of failure; if compromised, an attacker can bypass filtering rules and gain control over multiple networks.
 4. Powertrain, Chassis, Safety ECUs: Control critical driving and safety functions (engine, brakes, steering, ADAS).
 - Vulnerability: Direct manipulation by injecting forged CAN messages (after compromising a gateway), leading to control over critical functions like braking or acceleration.
 5. OBD-II Port: Physical port for diagnostics and maintenance.
 - Vulnerability: With physical access, an attacker can connect a device to inject malware, extract data, or reprogram ECUs.
 6. Sensor Systems: (Cameras, Radar, LiDAR) Provide environmental perception for ADAS and autonomous driving.
 - Vulnerability: Spoofing or jamming sensor signals to deceive the vehicle about objects, lanes, or traffic signs, causing incorrect driving decisions.
 7. Software Update Mechanisms (OTA): Allows remote software and firmware updates.
 - Vulnerability: Injection of malicious updates if the process isn't properly authenticated and encrypted, or integrity checks are bypassed.